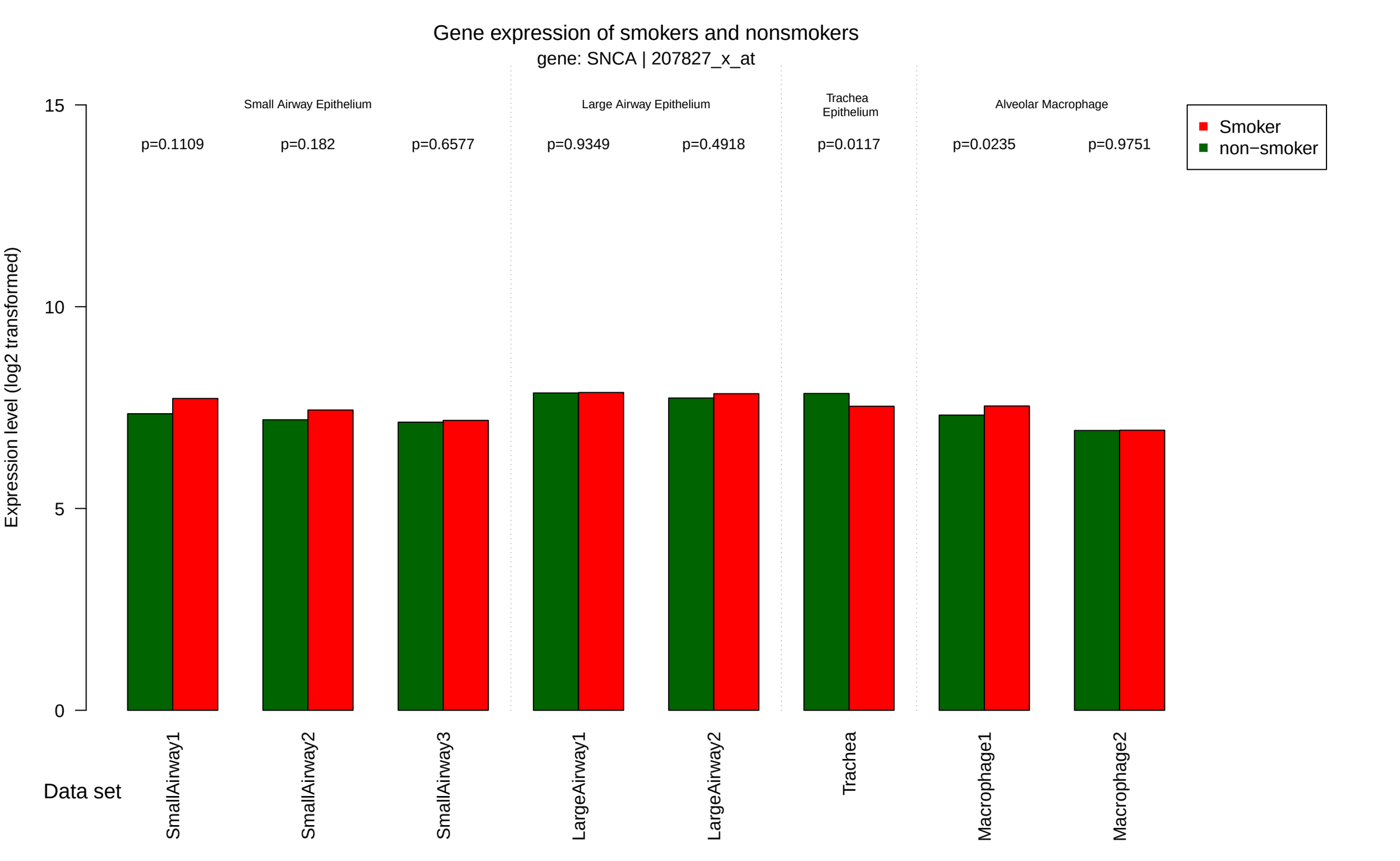
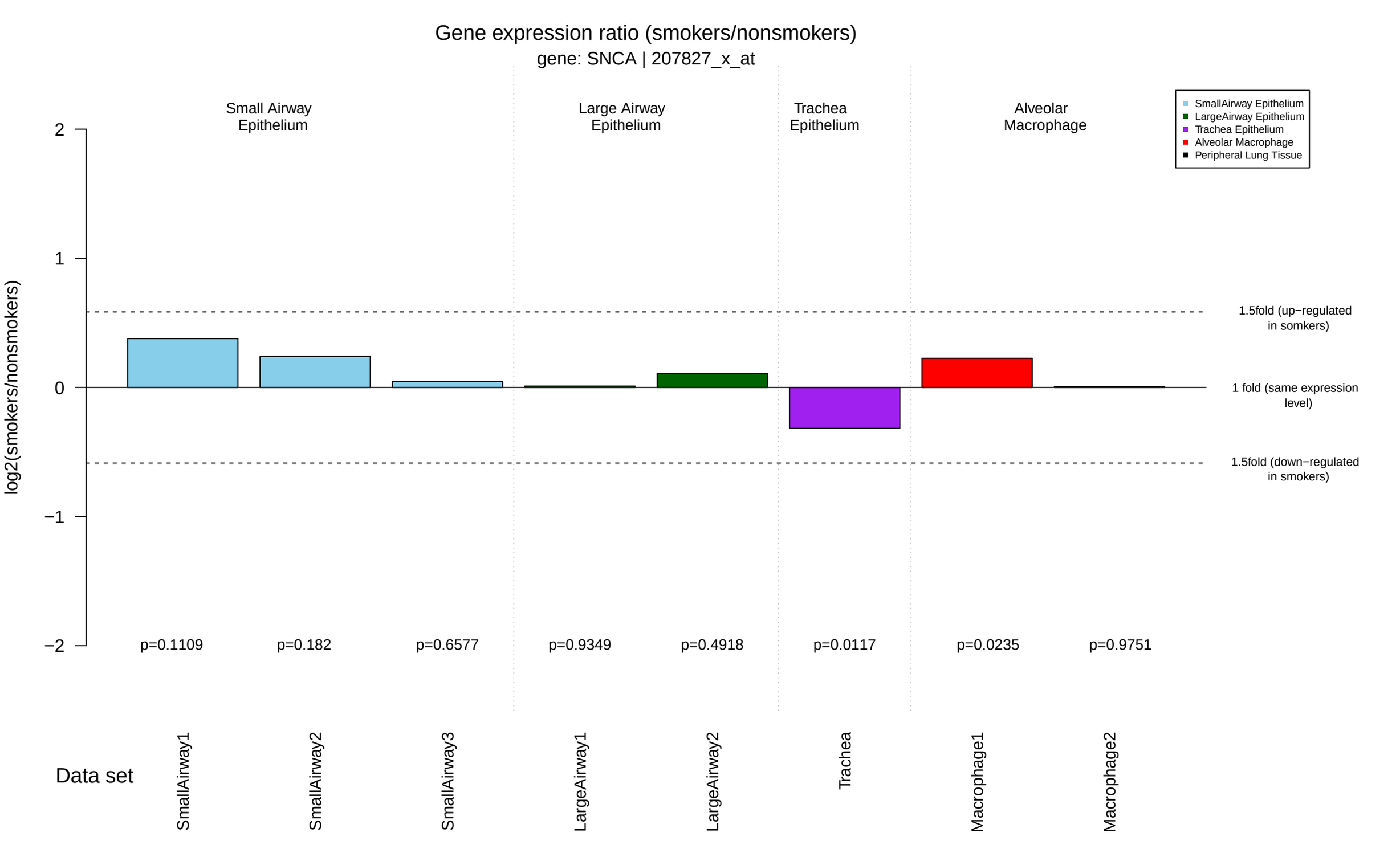
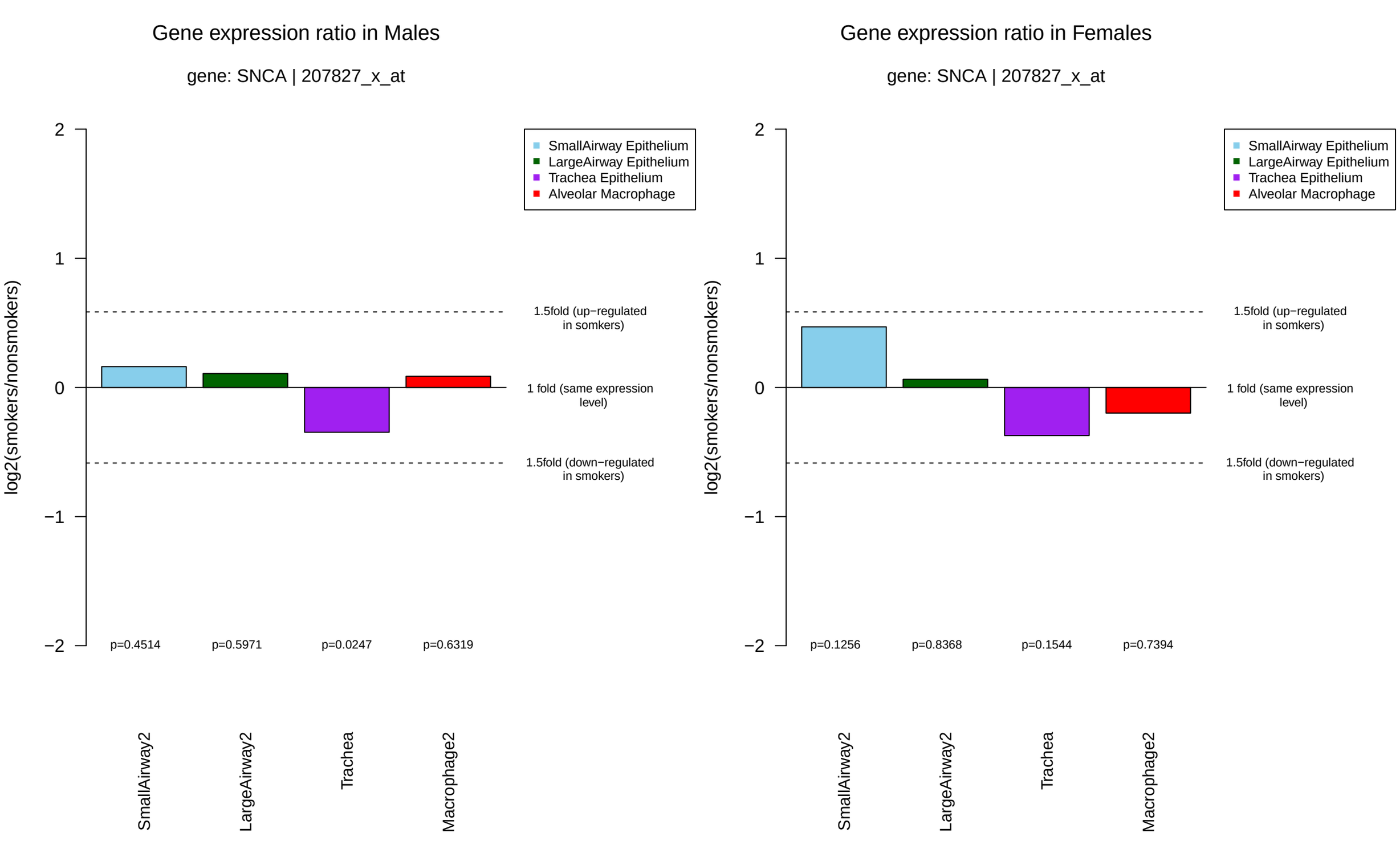
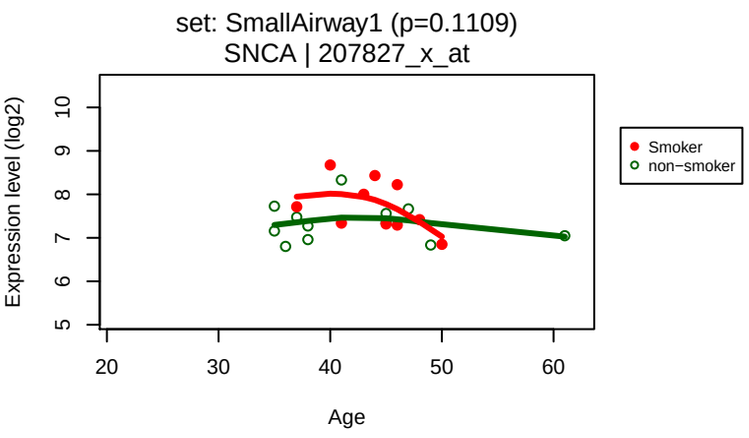
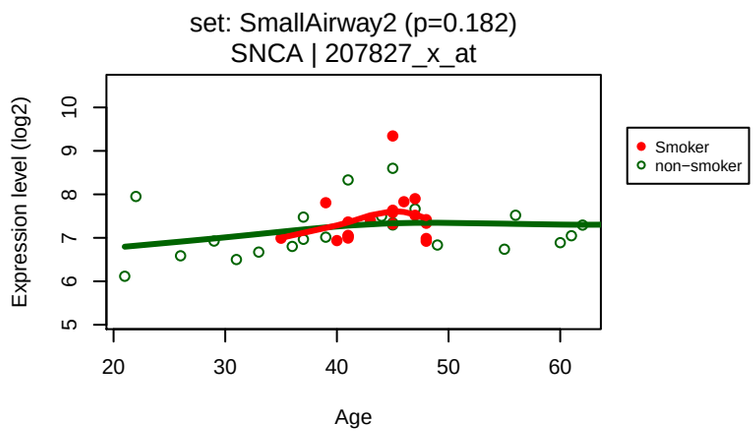
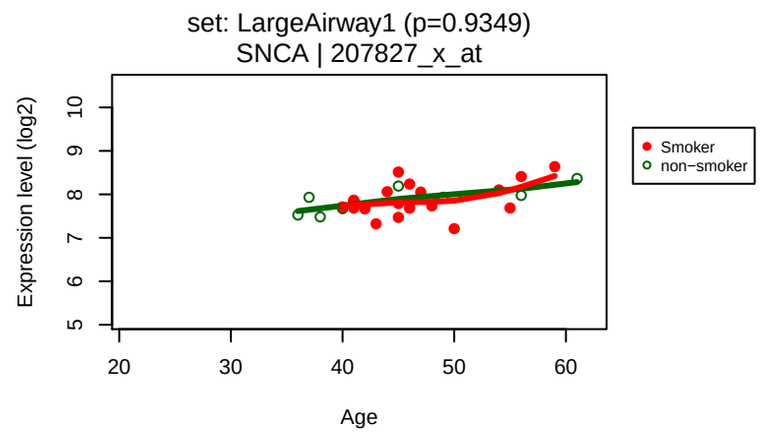

Gene expression of smokers and nonsmokers
gene: SNCA | 207827_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1109
p=0.182
p=0.6577
p=0.9349
p=0.4918
p=0.0117
p=0.0235
p=0.9751
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

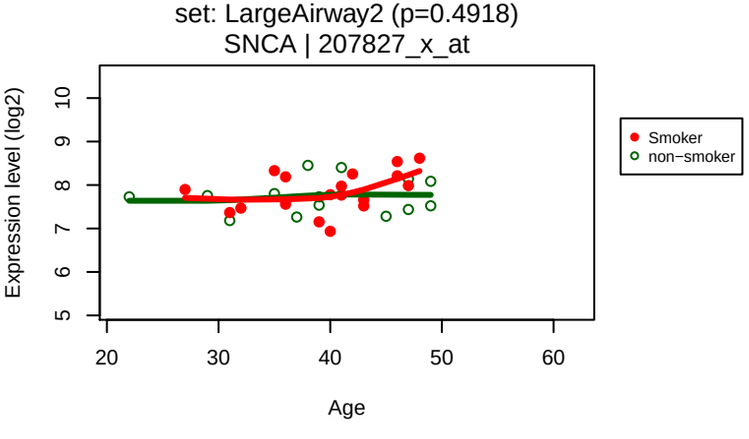

Gene expression ratio (smokers/nonsmokers)
gene: SNCA | 207827_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1109
p=0.182
p=0.6577
p=0.9349
p=0.4918
p=0.0117
p=0.0235
p=0.9751
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

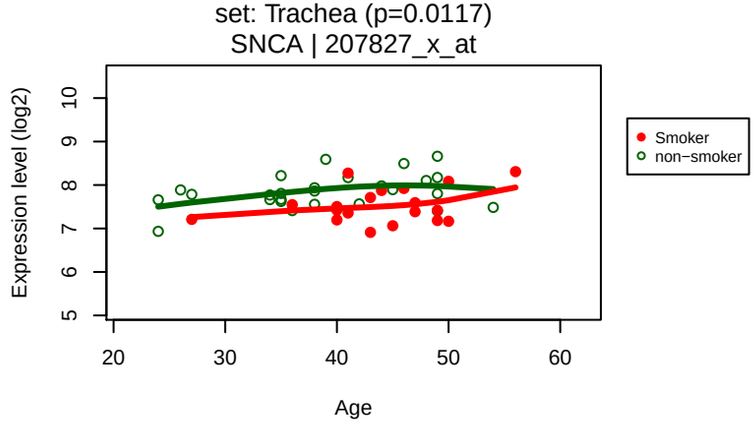

Gene expression ratio in Males
gene: SNCA | 207827_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4514
p=0.5971
p=0.0247
p=0.6319
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SNCA | 207827_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1256
p=0.8368
p=0.1544
p=0.7394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

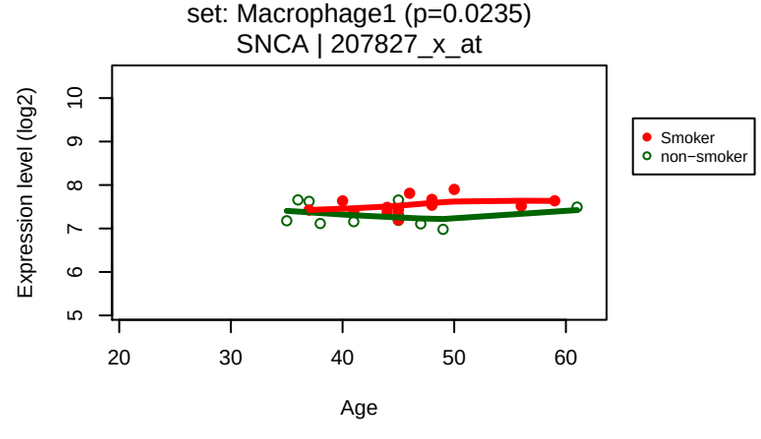

set: SmallAirway1 (p=0.1109)
SNCA | 207827_x_at
Expression level (log2)
Age
Smoker
non−smoker

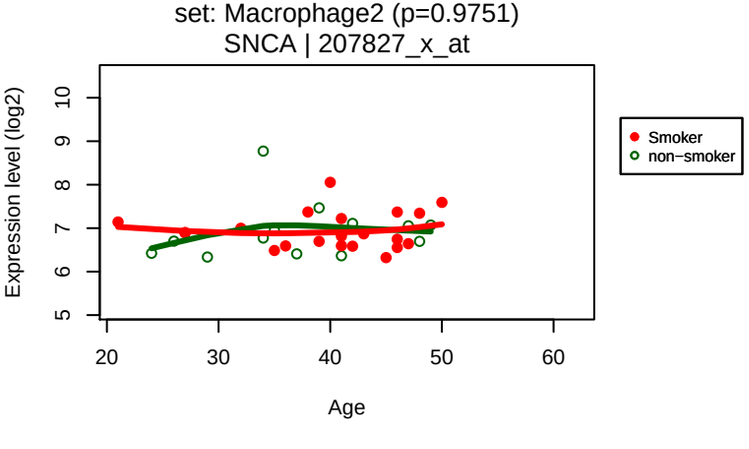

set: SmallAirway2 (p=0.182)
SNCA | 207827_x_at
Expression level (log2)
Age
Smoker
non−smoker

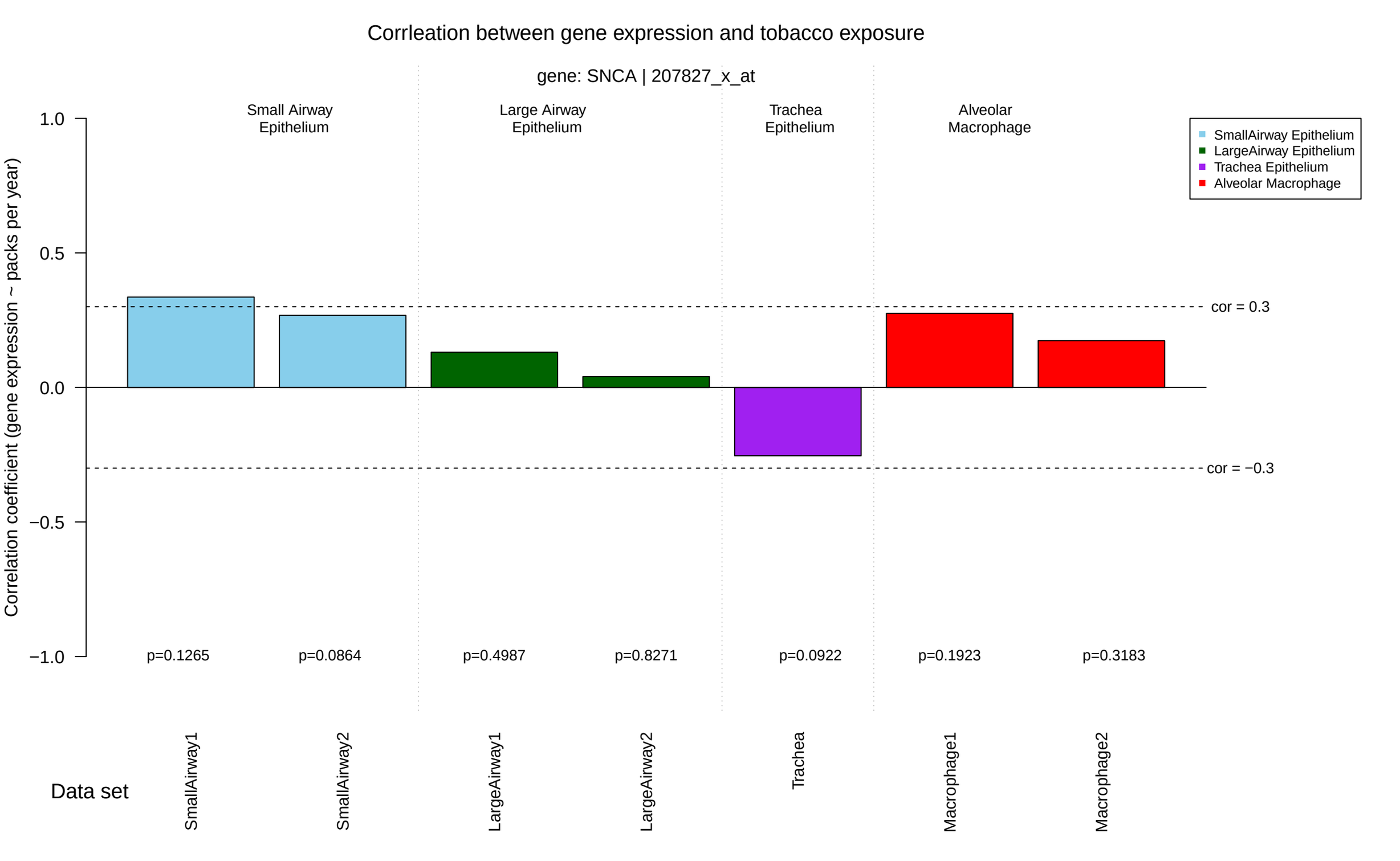

set: LargeAirway1 (p=0.9349)
SNCA | 207827_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4918)
SNCA | 207827_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0117)
SNCA | 207827_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0235)
SNCA | 207827_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9751)
SNCA | 207827_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SNCA | 207827_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1265
p=0.0864
p=0.4987
p=0.8271
p=0.0922
p=0.1923
p=0.3183
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2